Ethanoic Acid & Esterification Reactions

Exam code: 0620 & 0971

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Formation of ethanoic acid

Extended tier only

- Two methods used to make carboxylic acids are:
  - Oxidation by **fermentation**
  - Using **oxidising agents**

- The **microbial oxidation** (fermentation) of ethanol will produce a weak solution of vinegar (ethanoic acid)
- This occurs when a bottle of wine is opened as bacteria in the air (acetobacter) will use atmospheric oxygen from air to oxidise the ethanol in the wine

$$C_2H_5OH\,(aq) + O_2\,(g) \rightarrow CH_3COOH\,(aq) + H_2O\,(l)$$

- The acidic, vinegary taste of wine which has been left open for several days is due to the presence of ethanoic acid
- Alternatively, oxidising agent **potassium manganate(VII)** can be used
- This involves heating ethanol with **acidified potassium manganate(VII)** in the presence of an acid
- The heating is performed under **reflux** which involves heating the reaction mixture in a vessel with a condenser attached to the top
- The condenser prevents the volatile alcohol from escaping the reaction vessel as alcohols have low boiling points
- The equation for the reaction is:

$$CH_3CH_2OH\,(aq) + 2[O] \rightarrow CH_3COOH\,(aq) + H_2O\,(l)$$

- The solution will change from purple to colourless
- The oxidising agent is represented by the symbol for oxygen in square brackets

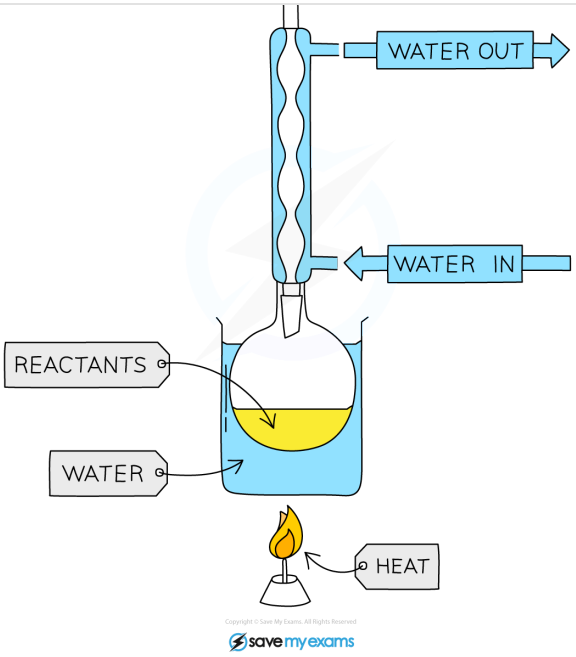
Diagram showing the experimental setup for the oxidation with $KMnO_4$ using reflux apparatus

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



Ethanol is heated with acidified potassium manganate(VII) in the presence of an acid

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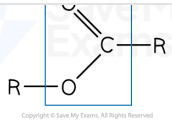
Esterification

Extended tier only

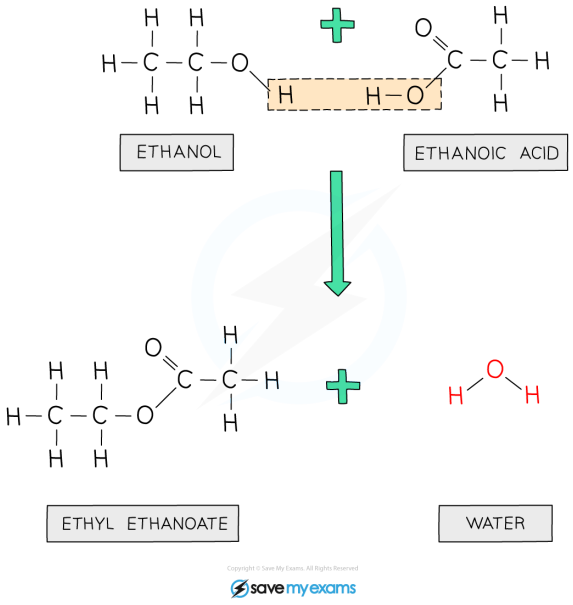
- Alcohols and carboxylic acids react to make esters in **esterification** reactions
- Esters are compounds with the functional group R-COO-R



- Esters are sweet-smelling oily liquids used in **food flavourings** and **perfumes**
- An acid catalyst is required for an esterification reaction
- Ethanoic acid will react with ethanol in the presence of concentrated sulfuric acid (catalyst) to form ethyl ethanoate:

$$CH_3COOH\,(aq) + C_2H_5OH\,(aq) \rightleftharpoons CH_3COOC_2H_5\,(aq) + H_2O\,(l)$$

Diagram showing the formation of ethyl ethanoate



During this esterification reaction, a molecule of water is also produced

Naming Esters

- An ester is made from an alcohol and carboxylic acid
- The first part of the name indicates the length of the carbon chain in the alcohol, and it ends with the letters '- yl'
- The second part of the name indicates the length of the carbon chain in the carboxylic acid, and it ends with the letters '- oate'
  - E.g. the ester formed from **pent**anol and **butan**oic acid is called **pent**yl **butan**oate

Diagram showing the origin of each carbon chain in ester; this ester is ethyl butanoate

Summary table of some esters and the compounds they are formed from

Name of ester	Alcohol and carboxylic acid it is formed from
propyl methanoate	propanol and methanoic acid
methyl butanoate	methanol and butanoic acid
ethyl propanoate	ethanol and propanoic acid

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